



EES404: Basic Meteorology
Midsem-II examination
IISER Mohali

Total Marks: 20

Time: 1 hour

PART-A: Short Answers

(4×2.5=10)

1. Perform the scale analysis for a tornado in the middle latitude, whose characteristic scales are: $U \sim 100 \text{ m/s}$; $W \sim 10 \text{ m/s}$; $L \sim 100 \text{ m}$; $\Delta P \sim 50 \text{ mb}$; $f \sim 10^{-4} \text{ s}^{-1}$; $H \sim 10 \text{ km}$. Consider vertical momentum equation here.
2. What are the Geostrophic and Gradient wind balances in the atmosphere? Derive their respective equations.
3. How is the general circulation of the atmosphere maintained? Explain with the help of a schematic diagram.
4. Explain the Clausius–Clapeyron equation and discuss its physical significance.

PART-B: Multiple Choice Questions or one word answers

(10×1=10)

1. An adiabatic process is one in which the
 - a) altitude of the air parcel remains constant
 - b) heat exchanged with the surroundings is zero
 - c) temperature remains constant
 - d) work done is zero
2. Thermal wind is a component of wind and blows parallel to the
3. The stability of an air layer refers to its
 - a) albedo
 - b) average temperature
 - c) pressure as measured at its base
 - d) tendency to either sustain or suppress vertical motions
4. The buoyancy of a rising air parcel is greatly affected by
 - a) wind speed
 - b) air pressure
 - c) heat released if water vapor condenses
 - d) absorption of solar radiation

5. Buys-Ballots' law states that, "In the northern hemisphere if you stand with your back to the surface wind, then turn clockwise about 30° , lower pressure will be:
- (A) To your right (B) To your left (C) Behind you (D) Ahead of you
6. An apparent force created by earth's rotation is the:
- (A) Pressure Gradient force (B) Centrifugal Force
(C) Coriolis Force (D) Viscous Force
7. A balance of Coriolis and Centrifugal forces:
8. Arrange the following horizontal scales of atmospheric motions in increasing order:
(A) small eddies, (b) planetary waves, (c) fronts, (d) synoptic cyclones
- A. fronts, planetary waves, synoptic cyclones, small eddies
B. small eddies, fronts, synoptic cyclones, planetary waves,
C. small eddies, synoptic cyclones, fronts, planetary waves,
D. small eddies, planetary waves, fronts, synoptic cyclones
9. For typical mean wind velocity in the tropics, the Rossby number is large. Choose the **best** answer.
- (A) True (B) False
10. During the day, if you stand on beach, what would be the wind direction ?
- A. From Sea to Beach
B. From Land to Sea
C. Within Sea
D. Along the beach shore